



saurav16997



Overview

Repositories 26

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Packages

Stars 15

**Saurav Premkumar**

saurav16997 · he/him



Focusing

Transforming data into insights | Bridging tech & business with AI-driven solutions | AI Engineer | Ex-BIE at SMEs | RA @ NYU Stern |

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saurav16997 / README.md



Saurav

AI Engineer | LLM Apps | RAG | MCP | Data Systems



About Me

Hi, I'm an AI/data engineer with 5 years of experience across analytics, BI, data systems, and AI-adjacent workflows, plus an M.S. in Information Systems from **NYU**.

I like building practical AI systems that connect language models with real tools, data, APIs, and business workflows. My recent work focuses on **LLM applications, RAG, MCP servers, agentic tooling, data pipelines, analytics engineering, and AI-powered automation.**

I'm especially interested in the space where **data** gives structure to messy problems, and AI helps turn that structure into usable products.

What I'm Building Toward

- LLM applications that use tools, APIs, retrieval, and structured outputs
- RAG systems that combine private knowledge with reliable generation
- MCP/tool-calling systems that let models interact with real-world data sources
- AI automation workflows for research, transcription, job discovery, and business analysis
- Data products that connect analytics, experimentation, and decision-making

Technical Skills

AI Engineering & LLMs

OpenAI LLM Apps RAG Model Context Protocol Tool Calling AI Agents Prompt Engineering
Vector Search ChromaDB OpenAI Whisper

Data, Backend & Analytics

Python MySQL SQL FastAPI Pandas NumPy Scikit-learn PyTorch Tableau Power BI
dbt BigQuery AWS

Featured AI Engineering Projects

• IdeaGen — AI Agent Business Idea Generator

Next.js React TypeScript FastAPI Python OpenAI Clerk Vercel

Built a full-stack SaaS demo that generates pitch-ready AI-agent business ideas in real time. Users sign in, hit the app, and watch a structured startup concept stream in live — name, hook, customer, agent workflow, monetization, unfair advantage, and a 30-day launch sprint — so visitors get a product experience, not a static chatbot dump.

- Ships a Next.js + Clerk frontend with authenticated access to the idea workspace
- Protects a FastAPI `/api` endpoint with Clerk JWT verification before calling OpenAI
- Streams completions over SSE so ideas appear token-by-token and render as live markdown
- Uses a custom IdeaSpark system prompt to keep outputs bold, specific, and founder-ready

[Live Demo](#) · [View Project](#)

• MediNotes Pro — AI Clinical Documentation Assistant



Built a full-stack healthcare SaaS demo that turns raw consultation notes into clinical documentation in real time. Doctors sign in, enter patient name, visit date, and notes, then watch AI stream a visit summary, next steps, and a patient-friendly email draft — a usable product flow, not a static chatbot dump.

- Ships a Next.js + Clerk frontend with an authenticated consultation workspace and date picker
- Protects a FastAPI `POST /api` endpoint with Clerk JWT verification and a Pydantic `visit` payload
- Streams OpenAI completions over SSE so documentation appears token-by-token as live markdown
- Uses a clinical system prompt that returns three clear sections: doctor summary, next steps, and patient email

[Live Demo](#) · [View Project](#)

• AI Marketing Spend Analyst



Built an AI marketing analyst that lets operators ask plain-English questions about campaign spend and performance. The agent inspects the uploaded dataset, writes read-only SQL, runs it safely, and returns a business-style answer with assumptions, caveats, and evidence citations — so decisions are transparent, not black-box guesses.

- Turns campaign CSV data into an interactive analyst for marketing ops
- Writes and executes read-only SQL for ROAS, CPA, CTR, CVR, and channel rollups
- Wraps results into clear biz-style recommendations with assumptions and caveats
- Cites every important claim with evidence IDs so answers stay data-backed and auditable

[View Project](#)

• AI Digital Twin



Built a deployed AI twin that answers recruiter-style questions about my background, projects, experience, and career context. The goal was to turn my profile into an interactive assistant that can handle common screening conversations and notify me when it receives a question it cannot answer.

- Answers questions using my own career and project context
- Designed for recruiter-style interactions and portfolio discovery
- Sends a notification when the AI encounters an unknown question
- Turns a static profile into an always-available AI interface

[View Project](#)

• Multi-Agent Deep Research

OpenAI Agents SDK Python Gradio AI Agents

Built a multi-agent research system that takes a user question, plans the research process, runs searches, synthesizes findings, and produces a structured long-form report. The project mimics a frontier-style deep research workflow using specialized agents for planning, searching, writing, and delivery.

- Uses planner, search, writer, and email agents
- Breaks open-ended research questions into executable subtasks
- Synthesizes search results into structured research reports
- Supports end-to-end research automation from question to final output

[View Project](#)

• Autonomous Engineering Team using CrewAI

CrewAI OpenAI Python Gradio Docker MCP

Built an autonomous AI engineering team using CrewAI that takes a plain-English product brief and turns it into a working full-stack application. Four agents collaborate like a small software team: an Engineering Lead designs the system, a Backend Engineer writes the core Python logic, a Frontend Engineer builds a Gradio UI, and a Test Engineer writes and runs unit tests until the application passes.

- Uses four role-based CrewAI agents across design, backend, frontend, and QA
- Generates a trading account simulator with deposits, withdrawals, share trading, portfolio valuation, P/L, holdings, and transaction history
- Produces backend code, a Gradio frontend, architecture docs, tests, and a QA summary inside a shared sandbox
- Runs agent-written code in an isolated Docker + uv environment for safer execution
- Integrates Context7 MCP so agents can reference current Gradio documentation while building the UI

[View Project](#)

• Automated Sales Agentic Coordination

OpenAI Agents SDK Python Agent Handoffs Sales Automation

Built an agentic outbound-sales workflow that coordinates multiple AI agents to research prospects, generate tailored outreach, and manage role-specific handoffs. The project demonstrates how AI agents can automate parts of a GTM workflow while preserving structure, context, and control.

- Uses agent coordination and handoffs for sales-email generation
- Automates prospect research and tailored message creation
- Demonstrates multi-step workflow orchestration with LLM agents
- Built around a fictional compliance product, ComplAI

[View Project](#)

• CrewAI Multi-Agent Workflow

CrewAI Python AI Agents Workflow Automation

Built a CrewAI-based multi-agent workflow where specialized agents collaborate on a task, divide responsibilities, pass context between roles, and produce a structured final output. The project focuses on practical agent orchestration rather than a single-prompt demo.

- Defines role-based agents with clear responsibilities
- Coordinates multi-step task execution through CrewAI
- Produces structured outputs from collaborative agent workflows
- Demonstrates how agent teams can automate research, analysis, or business tasks

[View Project](#)

• Pharma MCP Server

Python MCP FastMCP openFDA

Built a FastMCP server that exposes FDA pharmaceutical data to MCP-compatible LLM clients. The server provides tools for official FDA drug labels and adverse event reports, allowing an LLM to route user questions to the right clinical data source.

- Exposes `lookup_drug` and `get_drug_adverse_events` tools through MCP
- Connects LLM clients to FDA label and FAERS adverse-event data
- Demonstrates tool schemas, docstring-driven routing, and agent-style tool use

[View Project](#)

• AI Brochure Generator

OpenAI Python Web Scraping LLM Workflow

Built an AI workflow that analyzes a company website, selects relevant pages, scrapes content, and generates a short company brochure in markdown using the OpenAI API.

- Scrapes website links and page content
- Uses an LLM to identify relevant pages
- Generates structured markdown collateral from unstructured website data

[View Project](#)

• YouTube Playlist Transcription with OpenAI Whisper

OpenAI Whisper Python Automation

Built a workflow to transcribe an entire YouTube playlist using OpenAI Whisper and store each video transcript for downstream analysis or content workflows.

- Automates transcription across multiple videos
- Produces structured transcript outputs
- Useful for summarization, search, research, and content intelligence workflows

[View Project](#)

Data & Analytics Projects

• Marketing Funnel Analysis & A/B Testing

Python Pandas SciPy A/B Testing

Analyzed eCommerce event-level behavioral data to diagnose funnel drop-offs and evaluate a simulated A/B test.

- Built session-level funnel metrics across View → Cart → Purchase
- Identified View → Cart as the primary conversion bottleneck
- Evaluated experiment results using a two-proportion z-test and confidence intervals
- Translated statistical output into a business recommendation

[View Project](#)

• Data-Driven Insights into Consumer Behavior and Sales

Python Exploratory Data Analysis Marketing Analytics

Performed exploratory analysis on consumer behavior and sales data to identify marketing and sales insights.

- Analyzed customer behavior and sales patterns
- Focused on marketing-oriented business interpretation
- Developed recommendations for improving sales performance

[View Project](#)

How I Think About AI Systems

I like building AI systems that are useful beyond a demo:

User need
↓
Data / documents / APIs
↓
Retrieval, tools, or structured workflows
↓
LLM reasoning or generation
↓
Evaluation, automation, or business action



Pinned

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Data-Driven-Insights-into-Consumer-Behavior-and-Sales

Public

Analyze Amazon's Consumer Behavior and Sales datasets, conducting a robust Exploratory Data Analysis (EDA) with a focus on Marketing insights. Extracted actionable insights, offering strategic path...

☐ Jupyter Notebook  1



Predicting-Breast-Cancer-using-Data-Mining

Public

☐ Jupyter Notebook  1



Data-Analysis-for-Business

Public

Transcription-using-OpenAI-Whisper

Public

How to transcribe an entire YouTube playlist and store each video in cloud

☐ Jupyter Notebook  1



marketing-funnel-abtest-analysis

Public

Marketing Analyst and simulated AB testing

☐ Jupyter Notebook  3

 [saas-pricing-analytics](#) Public

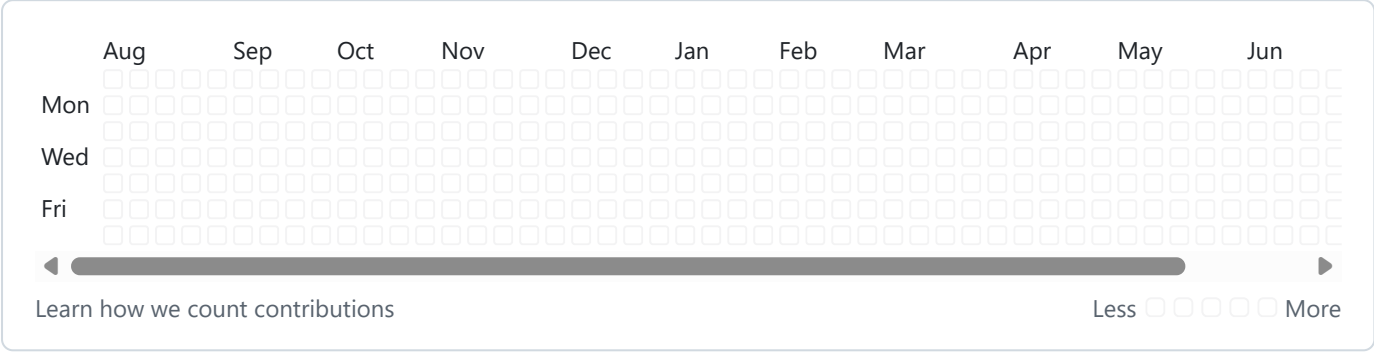
SaaS pricing intelligence and user behavior analytics (Notion-style)

☐ Python

 1

65 contributions in the last year

Contribution settings ▾



Contribution activity

Year: 2026 ▾

August 2026

- 

Created 12 commits in 3 repositories

[saurav16997/MediNotes-Pro-AI-Clinical...](#)
5 commits

[saurav16997/IdeaGen-AI-Agent-Business...](#)
4 commits

[saurav16997/saurav16997](#)
3 commits
- 

Created 2 repositories



[saurav16997/MediNotes-Pro-AI-Clinical-Docu...](#)

☐ TypeScript •

Built by  

Aug 5



[saurav16997/IdeaGen-AI-Agent-Business-Idea...](#)

☐ TypeScript •

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Aug 5
- Show more activity
- Seeing something unexpected? Take a look at the [GitHub profile guide](#).
- https://github.com/saurav16997
- 8/8